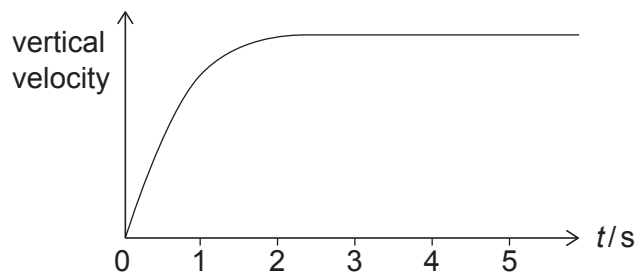


PAPER 2 / QUESTION 3

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3. A spherical oil droplet is released from rest at the bottom of a column of water.

The graph shows the variation with time t of the vertical velocity of the oil droplet.



- (a) The following data are available:

radius of the oil droplet = 3.5 mm

weight of the oil droplet = 1.6×10^{-3} N

density of the water = 1000 kg m^{-3}

viscosity of the water = 1.1×10^{-3} Pa s

- (i) Show that the volume of the oil droplet is about $2 \times 10^{-7} \text{ m}^3$. [1]

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- (ii) Calculate the initial acceleration of the oil droplet. [3]

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(Question 3 continued)

- (b) Describe why the acceleration of the oil droplet changes. [2]

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- (c) (i) Explain why the velocity of the oil droplet is constant for $t > 3$ s. [2]

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- (ii) Deduce the velocity of the oil droplet for $t > 3$ s. [2]

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